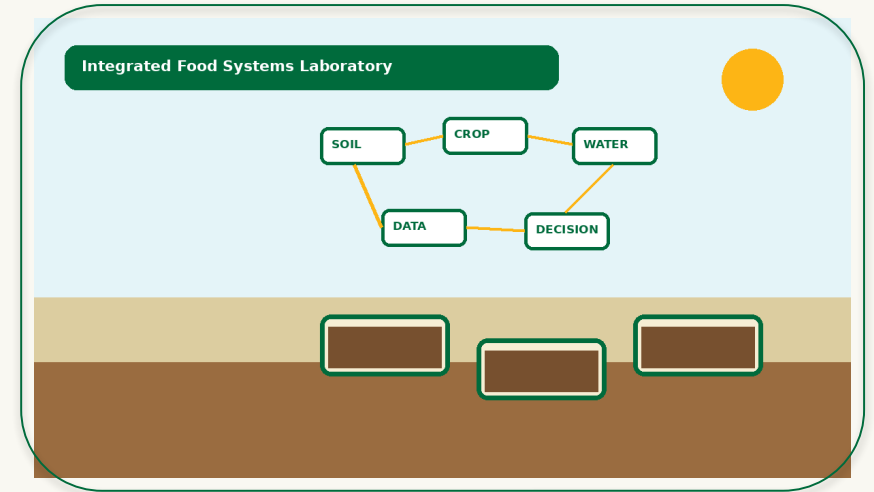


Integrated Food Systems Laboratory

Systems Analysis and Decision-Making

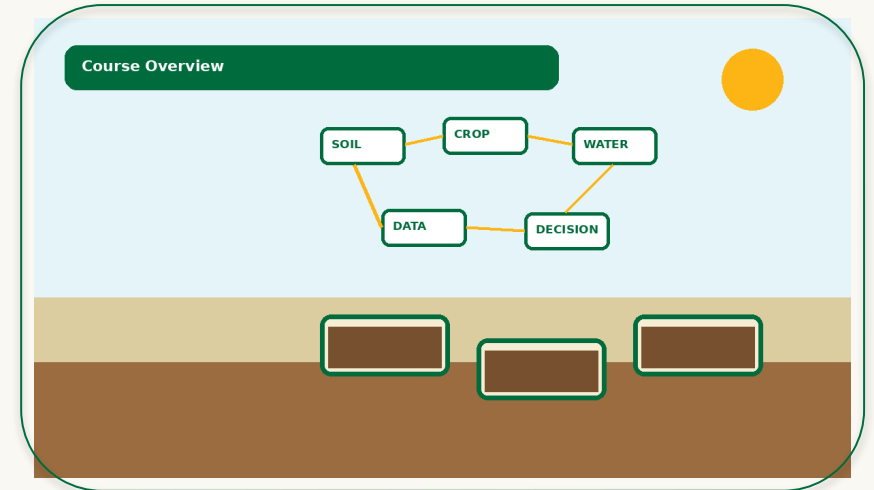
- LIFEWS AgriShine™ GrowMeal™ SSS 1-3 Teacher Training
- Audience: SSS 1-3 Agricultural Science, Biology, Mathematics and technical/STEM educators
- Format: Self-paced online course + supervised field/laboratory practical
- Recommended duration: 4 hours online + 2-3 hours practical
- SSS learning progression: design -> instrument -> model -> evaluate -> optimize -> innovate



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the system at senior-secondary technical depth; teach all ten modules accurately; supervise design, instrumentation, measurement and experimental work; maintain defensible technical records; guide quantitative analysis; distinguish measurement from inference; connect evidence to management, engineering and research decisions.
- Core materials: teacher guide, system drawings, approved tools, low-voltage educational hardware where relevant, measuring instruments, meters/sensors, data sheets/logger, clean sample/harvest containers, PPE/handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe technical practical, and submit an engineering design, monitoring plan, experimental protocol or systems-analysis report appropriate to the course.



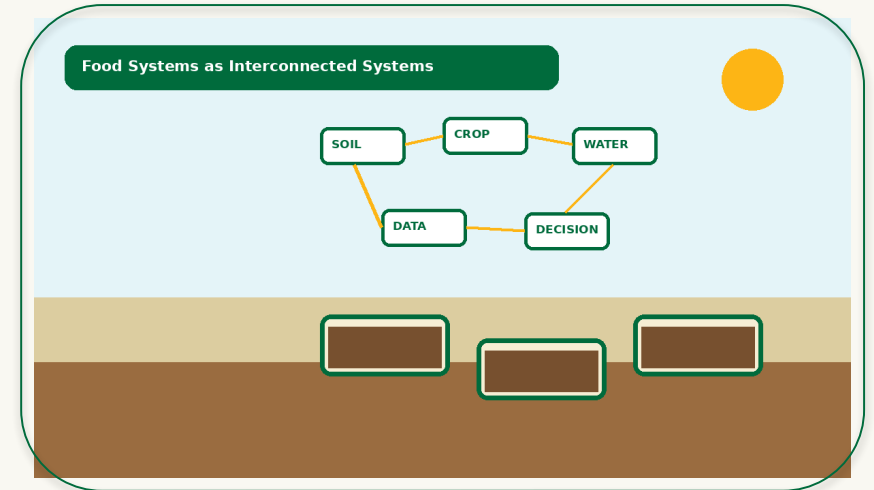
TEACHER PRACTICE

- Training sequence: DEFINE -> DESIGN -> IMPLEMENT -> MEASURE -> ANALYZE -> EVALUATE -> IMPROVE.

Food Systems as Interconnected Systems

Teacher technical knowledge, laboratory/field practice and student supervision

- Frame food production as interacting biological, physical, technological and human subsystems.
- Identify crops, soil, water, nutrients, infrastructure, energy, sensors, labor and decisions within the laboratory context.
- Use systems diagrams to represent connections and feedbacks.
- Recognize that improving one subsystem can affect others.
- Define the purpose of the system analysis before choosing metrics.



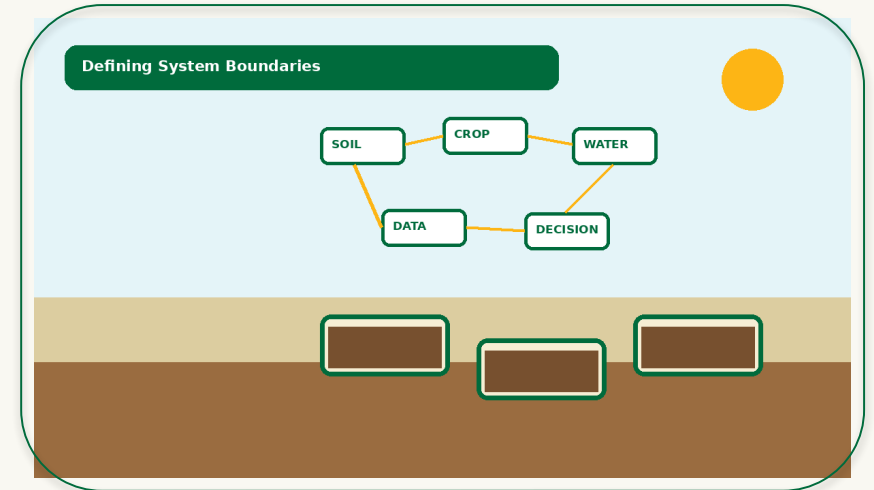
TEACHER PRACTICE

- Teacher task: create an integrated food-production system map.

Defining System Boundaries

Teacher technical knowledge, laboratory/field practice and student supervision

- Set spatial, temporal and functional boundaries appropriate to the question.
- Distinguish internal components from external inputs, outputs and influences.
- Explain how changing boundaries can change calculated performance.
- State whether analysis covers one bed, a production unit or the full school system.
- Document exclusions explicitly.



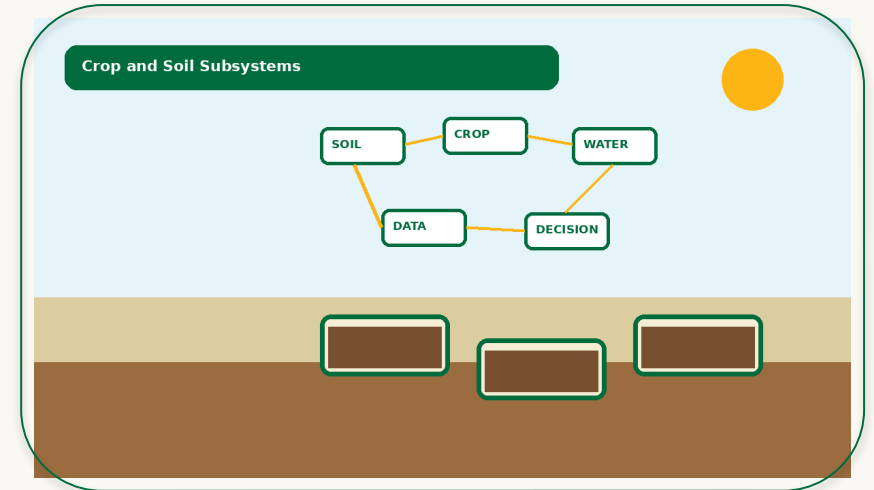
TEACHER PRACTICE

- Define two alternative boundaries for the same research question.

Crop and Soil Subsystems

Teacher technical knowledge, laboratory/field practice and student supervision

- Map crop growth, root activity, soil water, nutrients and organic matter interactions.
- Identify measurable state/performance variables.
- Connect soil management to crop outcomes using records and experiments.
- Recognize time lags between management action and crop response.
- Avoid reducing crop performance to a single soil factor.



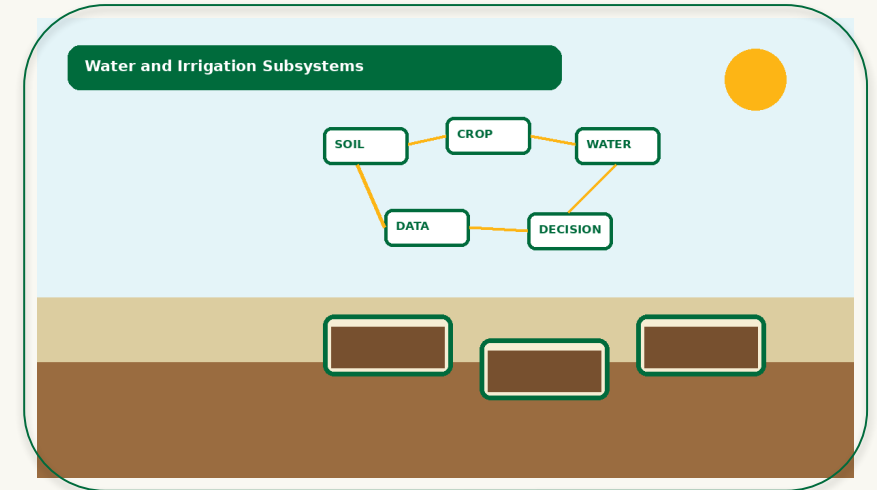
TEACHER PRACTICE

- Develop a crop-soil subsystem model.

Water and Irrigation Subsystems

Teacher technical knowledge, laboratory/field practice and student supervision

- Map source, storage, pumping/gravity, distribution, root-zone application, drainage and losses.
- Quantify flow, volume and storage where instruments permit.
- Connect irrigation decisions to crop stage and soil/root-zone measurements.
- Include reliability and maintenance constraints.
- Identify where water information enters management decisions.



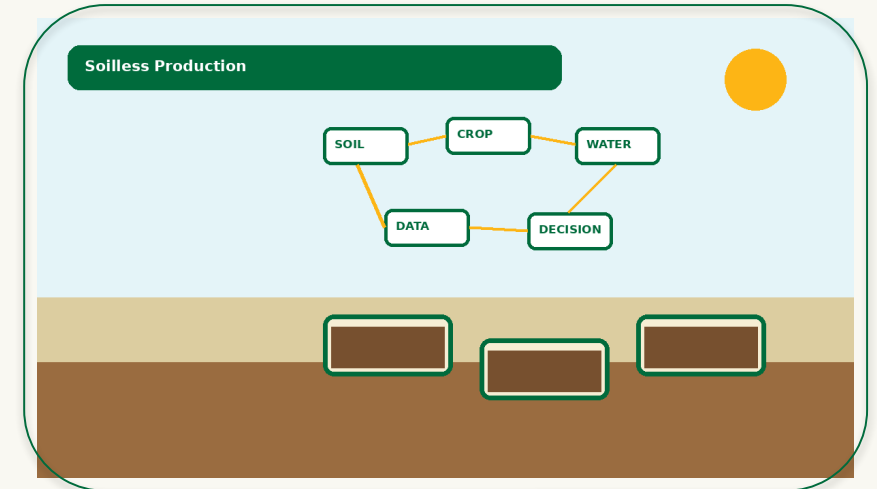
TEACHER PRACTICE

- Prepare a water-flow and decision diagram.

Soilless Production

Teacher technical knowledge, laboratory/field practice and student supervision

- Represent soilless systems as controlled water-nutrient-root-zone subsystems.
- Track water, nutrient solution, crop output and recirculation where applicable.
- Include pH, EC and system reliability as monitoring variables.
- Compare resource and management demands with soil-based production cautiously.
- Integrate the soilless unit into the broader laboratory boundary.



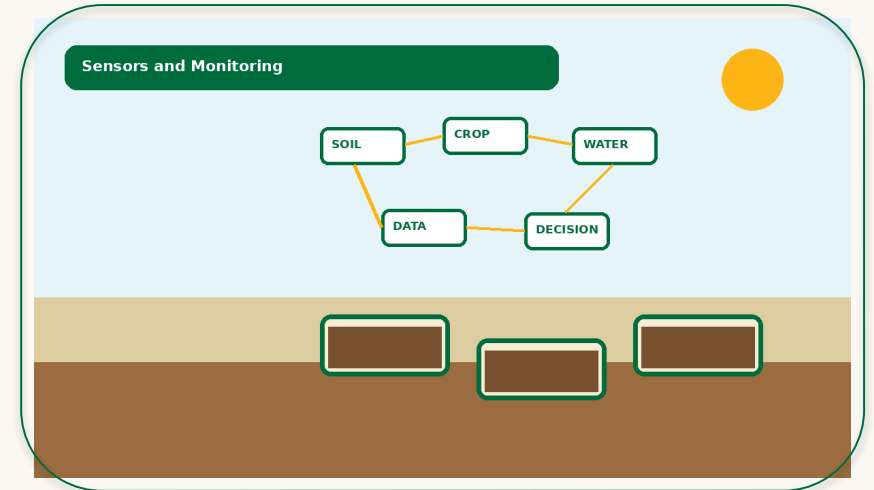
TEACHER PRACTICE

- Add the hydroponic subsystem to the system model.

Sensors and Monitoring

Teacher technical knowledge, laboratory/field practice and student supervision

- Map each sensor to the state or flow it measures.
- Define sampling frequency, units, data storage and validation requirements.
- Use monitoring data to update understanding of system condition.
- Identify blind spots where important variables are not measured.
- Keep measurement separate from automated decision authority unless control logic is explicitly defined.



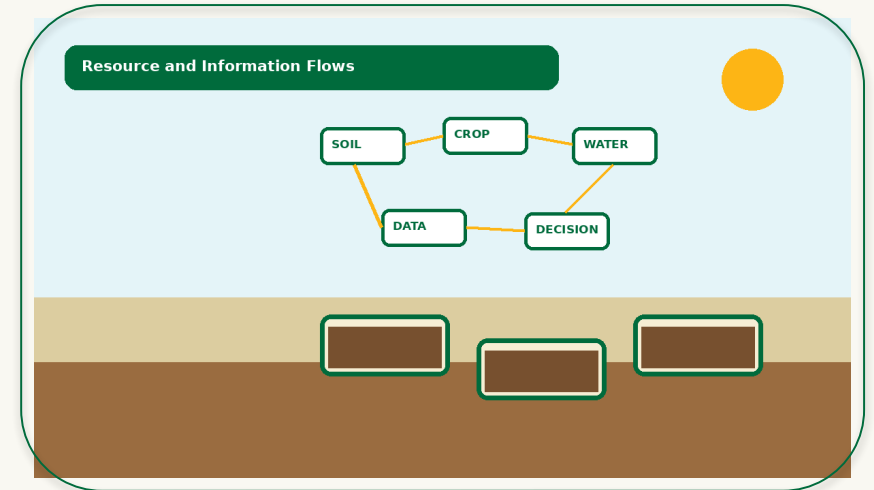
TEACHER PRACTICE

- Build a system monitoring matrix.

Resource and Information Flows

Teacher technical knowledge, laboratory/field practice and student supervision

- Trace water, nutrients, biomass/crops, energy, materials and information through the system.
- Distinguish stocks/storage from flows.
- Quantify selected flows using consistent units and time boundaries.
- Identify losses, reuse pathways and bottlenecks.
- Show information moving from sensors/records to decision-makers and controls.



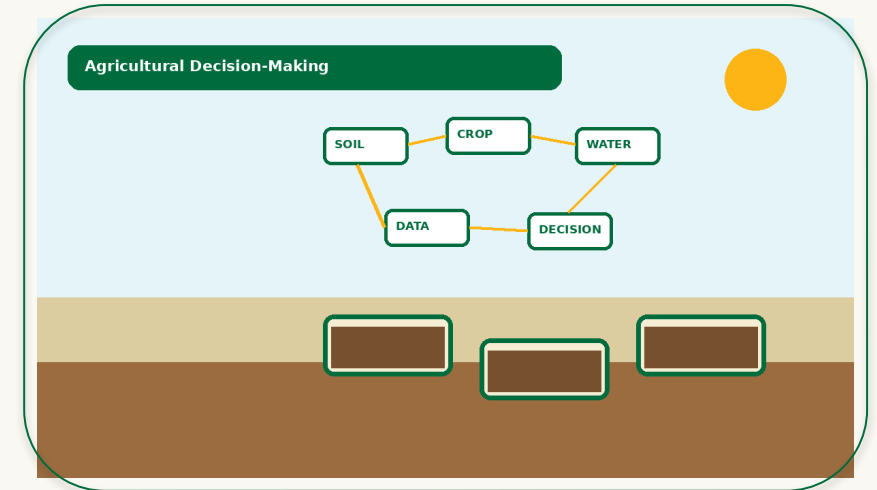
TEACHER PRACTICE

- Prepare a layered resource-and-information flow diagram.

Agricultural Decision-Making

Teacher technical knowledge, laboratory/field practice and student supervision

- Define management objectives and decision variables before evaluating options.
- Use measured data, constraints and expected consequences to compare alternatives.
- Distinguish operational decisions from longer-term design decisions.
- Include uncertainty and the cost of wrong decisions.
- Document the rationale for selected actions.



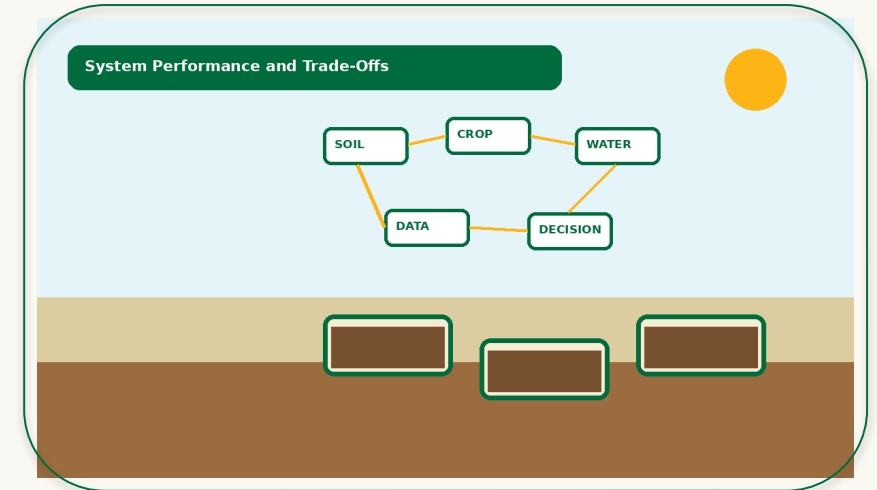
TEACHER PRACTICE

- Create a decision table for an irrigation or nutrient-management choice.

System Performance and Trade-Offs

Teacher technical knowledge, laboratory/field practice and student supervision

- Evaluate multiple metrics such as yield, water use, nutrient use, reliability, labor, energy demand, quality and cost where data exist.
- Recognize trade-offs when no option dominates every metric.
- Normalize or weight metrics only with transparent justification.
- Use scenario comparison rather than claiming one universal optimum.
- Identify sensitivity to key assumptions.



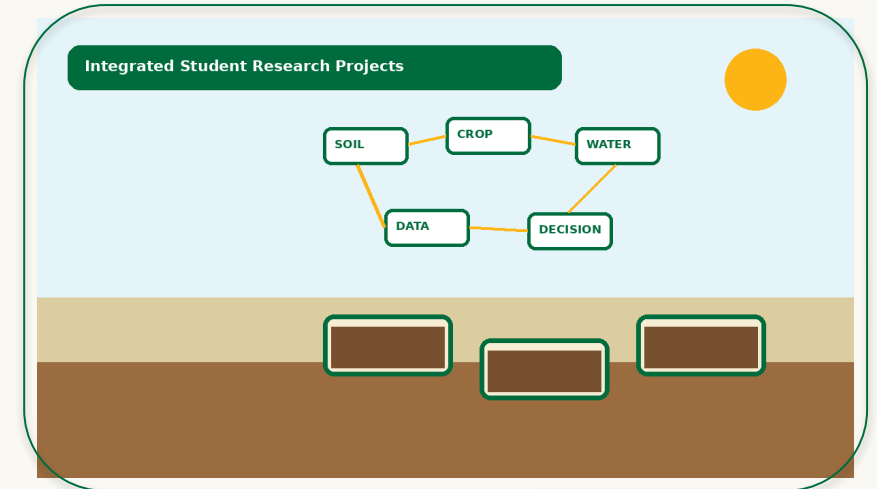
TEACHER PRACTICE

- Prepare a multi-criteria system performance comparison.

Integrated Student Research Projects

Teacher technical knowledge, laboratory/field practice and student supervision

- Require projects to connect at least two subsystems and define a system boundary.
- Develop research/design questions, measurements, data management and analysis before implementation.
- Use system diagrams to explain interactions.
- Evaluate management or design options using multiple performance indicators.
- Report limitations, uncertainty and recommendations for the next iteration.



TEACHER PRACTICE

- Completion task: supervise an integrated SSS research project from system map to decision.